Latent Epistemic Blocks:
Parallel Claim-Level Control for Retrieval and Verification

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Abstract

Retrieval need is not equivalent to model uncertainty, and a learned source classifier is unnecessary when transparent runtime routing closes its oracle gap. We pre-register a different experiment: can a parallel causal controller mark RETRIEVE, VERIFY, or HELP spans early enough to obtain or validate evidence before unsupported factual commitment? The latent head never predicts DB, lexical, vector, web, Iceberg, or another backend. Runtime owns evidence sufficiency, source resolution, credentials, provenance, status, and enforcement. We compare confidence, semantic rules, voluntary generic control, text and hidden-state sidecars, an independent watchdog, and oracle timing. This manuscript is a protocol, not a positive watchdog result; the Paper 2.5 source-router gate remains `no_go`.

1 Inherited Evidence and New Question

Self-RAG establishes learned on-demand retrieval and reflection in the main model [2]; learning whether to retrieve is not our novelty claim. FLARE provides a confidence-driven active-retrieval comparator [1]. Paper 1.5 indicates that epistemic requirement can differ from confidence and that runtime enforcement matters, but its public CRAG audit also shows that simple semantic rules can miss natural retrieval needs. Paper 2.5 finds no controlled heuristic-to-oracle source-routing gap. Therefore source-specific learning stays paused while claim timing and controller independence remain open.

The central question is: can a causal parallel controller mark the START, type, and END of an epistemic-risk span before commitment, without emitting control tokens, and does an independent watchdog rescue needs the generator's voluntary control misses?

2 Requirement, Availability, and Action

For candidate claim c_t, let $E(c_t)$ denote required evidence and A_t authoritative evidence already active in context, task state, or optional PRA-managed state. Retrieval action is not requirement alone:

$$\text{action}(c_t) = E(c_t) \setminus A_t.$$

After a RETRIEVE, VERIFY, or HELP block, the runtime (1) infers the evidence type; (2) checks active state; (3) reuses or materializes sufficient authoritative evidence; and (4) invokes Paper 2.5 R2 source routing only when evidence remains missing. The controller requests assistance; it does not assert truth.

3 Causal Claim-Level Controller

Paper 3's constrained state machine is reused. OUTSIDE predicts NONE or $\text{START}_{\{\text{retrieve,verify,help}\}}$; COMPUTE is a shared hard-negative/control type. INSIDE(type) predicts CONTINUE or END. Nesting and source identity are forbidden. On END, runtime checks evidence sufficiency, resolves a capability/source if needed, records typed evidence state, applies support policy, and resumes generation.

Every trajectory annotates earliest safe START s_e, latest acceptable START s_l, canonical END e, and first unsupported-value token u. A trigger is early when $s_e \leq s \leq s_l$, late when $s_l < s < u$, and missed-before-commitment when no valid trigger occurs before u. Lead time is $u - s$ tokens.

Table 1: The comparison separates voluntary control, independent observation, and source resolution.

Condition	Role
Paper 1.5 semantic runtime	Transparent epistemic-risk baseline
FLARE-like confidence	Confidence-only active retrieval
Paired/fenced/label-only intent	Voluntary generic language-bus control
Text-prefix sidecar	Model-independent learned observer
Final-layer latent controller	Frozen-LM hidden-state baseline
Multi-layer latent controller	Parallel depth-aggregated observer
Voluntary + watchdog hybrid	Tests independent rescue of voluntary misses
Self-RAG-style main model	Prior-art main-model comparator, faithfully labeled
Oracle timing/type	Upper bound before evidence and R2 errors

4 Trajectory Benchmark

Use generation trajectories, not only user questions. Include current, source-specific, private, or changed facts; attribution; unavailable, conflicting, and stale evidence; context/PRA-sufficient cases; quotations; hypotheticals and counterfactuals; user-given and computed facts; COMPUTE needs; and non-factual prose. Split paraphrase families, entities, relations, domains, and model families.

Each opportunity records intent, epistemic role, evidence-required and evidence-available flags, s_e, s_l, e, u, and authorized response policy. Roles include pending assertion, claim verification, uncertainty, quotation, hypothesis, counterfactual, user-given fact, computed/derived fact, and no retrieval. Source and query annotations support R2 diagnostics only and must not become R1 targets.

5 Matched Controllers

If exact Self-RAG training is infeasible, call the condition Self-RAG-inspired. Main-model training is confined to that declared comparator. Text and latent sidecars keep the generator frozen. Match controller complexity and runtime cost where possible; activation access alone earns no native-interface claim.

6 Independent Watchdog and Multi-Layer Control

Voluntary control may share the generator's blind spots. Compare voluntary-only, watchdog-only, and hybrid operation. Report error overlap and

$$\mathrm{WRR} = \frac{\text{voluntary misses caught safely by watchdog}}{\text{voluntary misses requiring intervention}}.$$

Watchdog Rescue Rate matters only alongside false activation, unnecessary retrieval, abstention cost, and lead time.

Reuse Paper 3's layer sweep and aggregation ladder: final layer, best development-selected layer, fixed logit averaging, voting, learned scalar weights, and a tiny MLP over logits. START and END may use different frozen mixtures. Plot decodability by depth and test whether RETRIEVE emerges earlier than VERIFY; probe accuracy alone is descriptive, not causal benefit.

7 R2 Resolution and Runtime Enforcement

R2 compares the existing Paper 2.5 heuristic, token-aware BM25, capability or source-native rules, broadcast, and oracle source. Learned source routing remains gated on a real heuristic-to-oracle gap. Registry/backend changes must not retrain R1.

Evidence statuses include supported, contradicted, unverified, stale, ambiguous, conflict, and unavailable. When required evidence is not authorized or sufficient, policy may force abstention, qualification, conflict reporting, or further retrieval. Measure evidence override when generation contradicts authoritative active state. An optional evidence-integration LoRA may be trained only after trigger quality is established and must remain factorized from the watchdog.

8 Metrics and Cost Frontier

Report START/END/type metrics and FAR; Early and Late Catch rates; missed-before-commitment; lead tokens; WRR and error correlation; unsupported commitment rate (UCR); Authorized Commitment Coverage; unnecessary retrieval; evidence override; abstention precision/recall; source/R2 correctness; calls, fanout, latency, and cost; calibration; and OOD/cross-model transfer. For controller threshold τ, report frontiers under

$$C(\tau) = \lambda_{fp} C_{fp} + \lambda_{fn} C_{fn} + \lambda_{lat} C_{lat} + \lambda_{tok} C_{tok},$$

not F1 alone. Mandatory slices include high-confidence/high-risk and low-confidence/low-risk trajectories, cases with evidence already available, conflicts, and each epistemic role.

9 Gates, Falsification, and Scope

The source-specific learned router is `no_go` while Paper 2.5's heuristic closes the oracle gap. The generic latent watchdog is an exploratory protocol because it tests timing and independent rescue, not source identity. It advances only if it improves the UCR–cost frontier over semantic rules and voluntary generic control with acceptable FAR and transfer.

Latent control is not justified if semantic rules match the frontier, voluntary intent catches nearly all needs, watchdog errors correlate with generator misses, false retrieval/abstention costs dominate rescues, or hidden states do not transfer. Main-model, text-sidecar, latent, hybrid, and task-dependent winners are all valid outcomes.

There is no transactional/backtracking contribution, learned source identity in R1, silent promotion of uncertain interpretation, or PRA/native-KV dependency. PRA may optionally search active task state before remote retrieval while leaving unrelated old records unmaterialized. Paper 4 proceeds only if retained Paper 3/3.5 state makes provenance, dependency, retraction, or transactional scope failures concrete.

10 Execution Order

Build the claim-level timing freeze from Paper 1.5 trajectories; reuse Paper 3 control code; run semantic and FLARE baselines; train the final-layer probe; sweep and aggregate layers; compare voluntary, watchdog, and hybrid control; add active-evidence sufficiency and enforcement; test cross-model/OOD transfer; and only then consider integration LoRA. Until those artifacts exist, this is a registered plan rather than evidence of an epistemic watchdog advantage.

References

[1] Z. Jiang et al. Active Retrieval Augmented Generation. *EMNLP*, 2023. `https://arxiv.org/abs/2305.06983`.

[2] A. Asai et al. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *ICLR*, 2024. `https://arxiv.org/abs/2310.11511`.